Report R22 — what the transition graph means

Weak components, terminal components and Hilbert-space fragmentation in open quantum systems

Tier 1 analysis

2026-08-18

Abstract

Every sector count this project reports is a count of components of one directed graph: the support graph of the channel in the computational basis. For unitary rules the choice of component notion is immaterial, and the project has been able to say "Krylov sector" without qualification. For dissipative rules it is not immaterial, and the two natural counts can disagree *exponentially*: we have rules with a single weakly connected component and ψ^N terminal components, $\psi = 1.4656$. This report says what the two counts are in the language of open quantum systems, and answers the question that provoked it — if the recurrent classes proliferate but the weak components do not, is that Hilbert-space fragmentation? The answer is **no**, and the reason is sharp rather than terminological. Weak components are exactly the blocks of a *strong symmetry* in the sense of Buča and Prosen: the finest decomposition of the basis that makes every Kraus operator block-diagonal, hence a genuine superselection rule with a charge measurable at $t = 0$. Terminal components are the *enclosures* of the Baumgartner–Narnhofer / Albert–Jiang asymptotic decomposition: they count steady states, not sectors. Fragmentation is a statement about the first; exponentially many attractors inside one block is multistability — an exponentially degenerate steady-state manifold with no symmetry protecting it. The distinction is operational, not aesthetic, and we measure it: the conserved quantities of a channel are the absorption probabilities $\varphi_i(x)$, and a strong symmetry is precisely the statement that they are **idempotent**. For rules whose weak components proliferate the initial state fixes the attractor with certainty for 100% of basis states; for rules with one weak component and exponentially many attractors it does so for 46% at $N = 10$, falling to 39% at $N = 12$ — the rest of the time the noise decides, not the initial condition. We map the four growth regimes onto the literature, flag a terminology collision between two independent "strong/weak" dichotomies that this project uses both of, and state precisely what a population-level graph can and cannot certify.

Contents

1 Why the question does not arise for closed systems

The object under study is the support graph G of one full brick-wall cycle Φ in the computational basis: vertices are the 2^N bit strings, and $x \to y$ when the channel can take $|x\rangle\langle x|$ to something with weight on $|y\rangle\langle y|$. Writing $\Phi(\rho) = \sum_j K_j \rho K_j^\dagger$, this is

$$x \to y \quad \Longleftrightarrow \quad \exists j : \langle y|K_j|x\rangle \neq 0. \tag{1}$$

Three notions of "component" live on a directed graph: weakly connected components (WCCs, obtained by forgetting edge directions), strongly connected components (SCCs, mutual reachability), and terminal SCCs — those with no outgoing edge, which we also call recurrent classes or attractors. In general these are different. The following says why the distinction has never had to be made in the closed-system fragmentation literature.

Proposition 1 (Unitarity collapses the three notions). *If $\Phi(\rho) = U\rho U^\dagger$ with U unitary, then every SCC of G is terminal and coincides with its WCC:*

$$\textit{WCC} \;=\; \textit{SCC} \;=\; \textit{terminal SCC}.$$

Proof. The induced population dynamics is the matrix $P_{yx} = |\langle y|U|x\rangle|^2$. Its columns sum to one because $U|x\rangle$ is normalised, and its rows sum to one because $\langle y|U$ is; so P is *doubly stochastic* and the uniform distribution is stationary with full support. In a finite Markov chain a transient state carries zero stationary mass, so no state is transient and every SCC is terminal. Finally, distinct terminal SCCs cannot be joined by an edge in either direction — an edge out of a terminal SCC would contradict terminality — so each is its own WCC. □

This is not an idle formality; it is why the project's earlier reports could speak of "the" Krylov sectors. Empirically it holds exactly: across all 16 unitary rules and all $N \in [6, 16]$ the measured transient fraction is identically 0 and $n_{\text{rec}} = n_{\text{wcc}}$ at every point, and no dissipative rule has an empty transient part at any N. The moment a reset channel D or E enters the rule word, P loses column normalisation on the right — two basis states are mapped onto one — the chain acquires transients, and the three counts separate. **The entire subject of this report is a dissipative phenomenon with no closed-system shadow.**

2 The dictionary

2.1 Weak components are a strong symmetry

Proposition 2 (WCCs block-diagonalise every Kraus operator). *Let $\{B_\alpha\}$ be the WCCs of G and P_α the orthogonal projector onto* $\operatorname{span}\{|x\rangle : x \in B_\alpha\}$*. Then $[K_j, P_\alpha] = 0$ for every Kraus operator K_j and every α, and consequently $\operatorname{Tr}(P_\alpha\rho)$ is conserved for all time. Conversely, the $\{B_\alpha\}$ are the* finest *partition of the computational basis with this property.*

Proof. If $P_\alpha K_j P_\beta \neq 0$ for $\alpha \neq \beta$ there are $x \in B_\beta$, $y \in B_\alpha$ with $\langle y|K_j|x\rangle \neq 0$, i.e. an edge $x \to y$ in (1); then x and y are weakly connected and $\alpha = \beta$, a contradiction. Hence each $K_j = \bigoplus_\alpha P_\alpha K_j P_\alpha$ and $P_\alpha K_j = K_j P_\alpha$. Conservation follows: $\operatorname{Tr}(P_\alpha \Phi(\rho)) = \sum_j \operatorname{Tr}(K_j P_\alpha \rho K_j^\dagger) =$

$\sum_j \mathrm{Tr}(K_j^\dagger K_j P_\alpha \rho) = \mathrm{Tr}(P_\alpha \rho)$, using $\sum_j K_j^\dagger K_j = \mathbb{1}$. Minimality holds because any coarser-than-G partition with block-diagonal Kraus operators has no edges between its parts, so it is a union of WCCs. □

Proposition 2 is the whole content of the identification. In the classification of Buča and Prosen [1], a symmetry of a Lindbladian is *strong* when the Hamiltonian *and every jump operator individually* commute with it, and *weak* when only the generator as a whole does. Setting $U = \sum_\alpha e^{i\theta_\alpha} P_\alpha$ for generic phases, Proposition 2 says exactly that the WCC decomposition furnishes a strong symmetry, and the minimality clause says it is the maximal one available in the computational basis. Buča and Prosen establish the fact that gives this report its answer: *only* a strong symmetry implies non-trivial reductions inside the space of steady states.

So:

> n_{wcc} counts superselection sectors. It is the open-system successor of "number of Krylov sectors", and it is the count that fragmentation is about.

2.2 Terminal components are enclosures

The terminal SCCs are a different kind of object entirely. They are the *enclosures* of Baumgartner and Narnhofer [2], whose analysis of GKS–Lindblad generators equips the Hilbert space with a decomposition into mutually orthogonal subspaces and separates the transient part from the part where probability accumulates. In the quantum-channel formulation the same structure is the irreducible decomposition of Carbone and Pautrat [4]: the recurrent subspace, which supports the invariant states, decomposes into the supports of the extremal invariant states, with the orthogonal complement being transient. Albert and Jiang [3] give the form of the asymptotic manifold,

$$\rho(\infty) = \bigoplus_k (\sigma_k \otimes \rho_k), \qquad \sigma_k \in M_{n_k}, \tag{2}$$

where k runs over the enclosures, ρ_k is a fixed state on the "dissipative" factor of enclosure k, and M_{n_k} is a noiseless subsystem carrying surviving coherence. Their central point — and the one we lean on in §3 — is that the infinite-time state is fixed by the initial state together with the full list of conserved quantities.

So:

> n_{rec} counts steady states. It is a statement about the $t \to \infty$ limit, not about a decomposition of the Hilbert space.

2.3 The table

graph object	algebraic object	physical reading
WCC	block of a strong symmetry; $[K_j, P_\alpha] = 0$	superselection sector; Krylov sector of the dephased model
n_{wcc}	# blocks	# conserved charges valid at *all* times
$d_{\max}^{\mathrm{wcc}}$	largest block dimension	the R21 strong/weak-fragmentation ratio
SCC	communicating class	refines the sector into transient strata
transient SCCs	transient subspace	everything that decays
terminal SCC	enclosure [2]; irreducible component [4]	attractor; support of one extremal steady state
n_{rec}	# enclosures	# extremal steady states
$d_{\max}^{\mathrm{rec}}$	$n_k \cdot m_k$ of the largest enclosure	noiseless-subsystem dimension *times* mixing dimension

Two structural facts follow immediately and are worth stating because they constrain every census. First, $n_{\rm rec} \geq n_{\rm wcc}$ always, since every weak component contains at least one terminal SCC; the inequality is saturated exactly when each sector has a unique attractor. Second, the familiar hyperbola $ab \geq 2$ of R9/R16 is a statement about WCCs *only*, and it holds because they tile: $\sum_\alpha \dim B_\alpha = 2^N$. Recurrent classes do not tile — they occupy the recurrent volume $\mu^N \leq 2^N$ — so the correct analogue is $a_{\rm rec}\, b_{\rm rec} \geq \mu$, with $\mu = 2$ recovering the closed-system bound. For the rules discussed below $\mu = \psi = 1.4656$ and the bound is saturated exactly.

3 Is exponential $n_{\rm rec}$ without exponential $n_{\rm wcc}$ fragmentation?

3.1 The criterion

The sharpest available definition of fragmentation is the commutant-algebra one of Moudgalya and Motrunich [5]: given the set of local terms (Hamiltonian terms, or the local gates of a circuit), form the algebra of all operators commuting with each of them; the system is fragmented when the dimension of this *commutant* grows exponentially in system size, as against the polynomial growth produced by conventional symmetries such as $U(1)$ or $SU(2)$. Their framework also supplies the classical/quantum distinction — "classical" fragmentation being fragmentation in a product basis, which is the case here.

The open-system version replaces "commutes with each local term" by "commutes with the Hamiltonian and each jump operator", which is Buča–Prosen strong symmetry, and the superoperator formulation is developed in [6], where symmetry algebras appear as frustration-free ground states of a local super-Hamiltonian and the strong/weak distinction is made explicit. Under that criterion the answer is immediate: Proposition 2 identifies the strong-symmetry commutant in the computational basis with the WCC decomposition, so **exponentially many weak components is fragmentation and exponentially many terminal components is not**. A rule with one weak component has a trivial strong-symmetry commutant: its Hilbert space is a single Krylov sector.

3.2 What is exponentially large instead

This does not make the phenomenon empty — something is exponentially large, and it is worth naming precisely. It is the space of conserved quantities in the Heisenberg picture. For each terminal component T_i define the *absorption probability*

$$\varphi_i(x) \;=\; \Pr[\,\text{absorbed in } T_i \mid \text{start in } x\,], \qquad J_i \;=\; \sum_x \varphi_i(x)\, |x\rangle\langle x|. \tag{3}$$

Each J_i satisfies $\Phi^\dagger(J_i) = J_i$, so $\operatorname{Tr}(J_i \rho(t))$ is constant: these are conserved quantities in exactly Albert and Jiang's sense, and there are $n_{\rm rec}$ of them. Exponentially many attractors therefore do imply exponentially many conserved quantities. The question is what *kind*.

Proposition 3 (Idempotency is the signature of a strong symmetry)**.** *$J_i^2 = J_i$ for every i if and only if $\varphi_i(x) \in \{0,1\}$ for every x and i, which holds if and only if every weak component contains exactly one terminal component, i.e. $n_{\rm rec} = n_{\rm wcc}$. In that case the J_i are the P_α of Proposition 2.*

This is the whole answer in one line. Fragmentation requires the conserved quantities to be *projectors* — charges that take a definite value on every basis state, that can be measured at $t = 0$, and whose value can never change. When $n_{\rm rec} \gg n_{\rm wcc}$ the conserved quantities exist but are not idempotent: they are non-local, real-valued functions on configuration space, they have no interpretation as a charge of the initial state, and they are recovered only in the $t \to \infty$ limit.

3.3 The operational content, measured

Proposition 3 converts the abstract distinction into a number one can compute: what fraction of basis states have their attractor determined *with certainty* by the initial condition? We built the honest population Markov chain of the channel (I keeps, $V = H$ splits $\frac{1}{2}/\frac{1}{2}$, D and E reset) and solved $\varphi = Q\varphi + R$ exactly on the transient block.

rule	regime	n_{wcc}	n_{rec}	certain fate	effective # attractors
W_{73} `VIID`	sectors proliferate	41	41	100.0%	1.00
W_{109} `EIIV`	sectors proliferate	54	54	100.0%	1.00
W_{36} `IDDV`	attractors only	10	60	72.5%	1.8
W_{203} `VEII`	attractors only	1	28	46.1%	2.6
W_{203} `VEII` ($N = 12$)	attractors only	1	60	39.1%	3.2

All rows at $N = 10$ unless marked, obc0. "Effective # attractors" is the participation ratio $1/\sum_i \varphi_i(x)^2$ averaged over basis states. The dissipative rules whose *weak* components proliferate behave exactly as Proposition 3 requires: every basis state has a certain fate, the conserved quantities are projectors, the charge is fixed at $t = 0$. W_{203}, with a single weak component, does not: fewer than half its basis states have a determined fate, and the fraction *falls* with N while the effective number of accessible attractors rises (max 15.0 at $N = 10$, 27.1 at $N = 12$).

The physical statement is therefore:

> With exponentially many weak components, **the initial state decides** which attractor the system reaches. With one weak component and exponentially many attractors, **the noise decides**; the initial state only sets the odds.

Both are ergodicity breaking, and both give an exponentially degenerate steady-state manifold. Only the first is Hilbert-space fragmentation.

3.4 Where this sits in the current literature

Two neighbouring bodies of work make the placement precise.

Dissipative freezing. Sánchez Muñoz, Buca and co-workers [10, 11] study what individual quantum trajectories do under a *strong* symmetry: initialised in a superposition of sectors, each stochastic realisation randomly selects one and stays there, so the conservation law that holds for the density matrix is "broken" at the level of single realisations. Our case is the mirror image. There, trajectory selection happens *despite* a strong symmetry that makes the ensemble remember; here there is no strong symmetry at all, and the trajectory-level selection among ψ^N attractors is the only structure present. The two phenomena are easy to confuse from the trajectory picture alone and are distinguished exactly by whether φ_i is idempotent.

Dark states. When the attractors are single basis states — the generic case among the rules where this gap appears — the terminal components are dark states in the standard sense: configurations immune to every jump operator, which is precisely how the recurrent subspace is characterised in open reaction–diffusion settings. An exponentially large dark-state manifold reached from a structureless transient bulk is a well-recognised target for dissipative state preparation, and it is a different resource from a fragmented Hilbert space: one prepares states, the other protects them.

4 The four regimes

Writing $n_{\text{wcc}} \sim a_{\text{w}}^N$, $n_{\text{rec}} \sim a_{\text{r}}^N$, $d_{\max}^{\text{rec}} \sim b_{\text{r}}^N$, four regimes are possible and each has a distinct reading.

(A/D) Sectors, attractors and attractor dimension all exponential. Exponentially many superselection sectors, each retaining an exponentially large recurrent space. This is fragmentation in the full sense, and it is the regime in which the asymptotic manifold (2) can carry an exponentially large noiseless subsystem — strong fragmentation is a known route to stabilising exponentially many decoherence-free subspaces. Whether $d_{\max}^{\mathrm{rec}} = n_k m_k$ is coherence (n_k) or classical mixing (m_k) is *not* decided by the graph; see §6. In our setting the unitary rules W_{108}, W_{201}, W_{156}, W_{198} sit here together with the dissipative W_{73}, W_{109}.

(A/C) Sectors exponential, attractors trivial. Exponentially many sectors, each collapsing onto a small attractor — typically one frozen state per sector. The fragmentation is real and the memory is classical: the long-time state is a bit string labelled by the conserved charge. This is what dephasing does to a fragmented model, and it is the regime Li, Sala and Pollmann [7] reach when they couple a fragmented system to a dephasing bath: the quantum fragmentation is reduced to a classical one, the stationary state becomes separable, and what survives is classical correlations arising from fluctuations of the remaining conserved quantities. The degenerate limiting case is $W_{204} = \mathtt{IIII}$, with 2^N sectors of one state each.

(B/C) Attractors exponential, sectors not, attractors trivial. The regime that provoked this report. One (or polynomially many) Krylov sectors, ψ^N dark states, a transient bulk occupying 99%+ of the Hilbert space, and non-idempotent conserved quantities. Not fragmentation: **multistability with an exponentially degenerate dark-state manifold**. The exemplar is $W_{203} = \mathtt{VEII}$, whose entire 2^N-dimensional space is one weak component.

(B/D) Attractors exponential and exponentially large, sectors not. Exponentially many enclosures each exponentially large, inside a single superselection sector. This would be the most interesting regime — an exponentially degenerate manifold of exponentially large noiseless subsystems with no symmetry organising them — and it is *empty* in the 256-rule family at obc0. We record its absence as an observation, not a theorem; nothing we know forbids it, and the volume constraint $a_{\mathrm{r}} b_{\mathrm{r}} \leq 2$ leaves room for it.

5 A terminology collision worth flagging

This project now uses two independent dichotomies that are both called "strong versus weak", and they are orthogonal. Conflating them would invalidate R21's conclusions or this report's, depending on the direction of the error.

strong / weak *fragmentation* Sala *et al.* [8], Khemani *et al.* [9]	**strong / weak *symmetry*** Buča–Prosen [1]
About the *size* of the largest sector relative to the symmetry sector containing it: strong iff $D_{\max}(s)/\dim s \to 0$.	About *how* a symmetry acts: strong iff every jump operator commutes with it individually, weak iff only the generator does.
A question you ask *after* you have a sector decomposition.	A question about whether you have a sector decomposition at all.
Used in R21, which grades 108/201/156/198 as strongly fragmented.	Used here, to say what n_{wcc} counts.

A model can be weakly fragmented with a perfectly good strong symmetry (W_{60}, W_{102} in R21), and — as §3 shows — can have exponentially many steady states with no strong symmetry whatever. There is also a third, newer use of the same words in the mixed-state literature: *strong-to-weak spontaneous symmetry breaking* [12], where a strong symmetry of a density matrix degrades to a weak one and the order parameter is a fidelity correlator rather than a two-point function. That is a statement about phases of mixed states, not about sector counting, and it should not be read into either column above. Whether the boundary between our regimes A and B can be crossed as a genuine SWSSB transition in a family of channels is a natural question and one we have not touched.

6 What a population graph cannot certify

The support graph (1) is the *population* skeleton of the channel. It records which basis states feed which, and nothing about the phases. Three consequences bound every claim in this report and in the project's dissipative reports generally.

1. **It is a lower bound on fragmentation.** A channel may have conserved quantities invisible in the computational basis — the fragmentation may be quantum, i.e. in a non-product basis, in the sense of [5]. The WCC count is then an undercount. This is not hypothetical for us: R-T15 identified the single-V commutant of $W_{108}/W_{201}/W_{156}/W_{198}$ as a Temperley–Lieb algebra, and the Temperley–Lieb model is exactly the vehicle Li, Sala and Pollmann use for *quantum* HSF in an entangled basis [7]. Their fragmentation-preserving coupling produces a steady state with coherent memory carried by an extensive set of *non-commuting* conserved quantities. No diagonal graph can see any of that.

2. **It cannot split $d^{\text{rec}}_{\max}$ into $n_k \cdot m_k$.** The graph gives the dimension of an enclosure's support; (2) says that dimension factorises into a noiseless subsystem and a mixing factor, and which one carries the growth is a question about coherence. The project has answered it in both directions already: R17 found the exponential terminal SCCs to be dense, mixing digraphs rather than cycles — so m_k is large and n_k small, a classical attractor — while the X-gate correction established the opposite for permutation circuits, where the functional graph describes only populations and exponentially large decoherence-free subspaces sit underneath it. *The graph alone never settles this.*

3. **The equality** $\dim(\textbf{fixed points of } \Phi^\dagger) = n_{\text{rec}}$ **is a classical statement.** For the population chain the conserved quantities are exhausted by the absorption probabilities (3). The full channel can carry additional fixed points built from coherences between enclosures of equal structure, which is precisely the M_{n_k} factor in (2). So n_{rec} is a lower bound on the steady-state count too.

7 Summary

- WCCs are the blocks of a *strong symmetry*: the finest basis decomposition making every Kraus operator block-diagonal (Prop. 2). Their count is the open-system Krylov sector count, and exponential growth of it is Hilbert-space fragmentation.

- Terminal SCCs are *enclosures*: supports of extremal steady states. Their count is a $t \to \infty$ statement and is always $\geq n_{\text{wcc}}$.

- Exponentially many terminal SCCs inside one WCC is **not** fragmentation. It is multistability: the conserved quantities are absorption probabilities, which fail to be idempotent (Prop. 3), and the attractor is selected by noise rather than fixed by the initial state — a 100% versus 46% effect in direct measurement (§3.3).

- The distinction has no closed-system analogue: unitarity makes the population dynamics doubly stochastic, which kills the transient part and collapses all three component notions (Prop. 1).

- "Strong/weak" means two different things in this project, and now a third thing in the mixed-state literature (§5).
- Everything here is certified at the level of populations only (§6); coherent fragmentation would be invisible, and our own Temperley–Lieb structure is a live candidate for it.

Reproduce

The measurement of §3.3 needs transition *probabilities*, which the project's component engine does not carry: `graph/scc.py` works with the unweighted successor relation, since that is all that component structure requires. The weighted one-cycle chain is therefore rebuilt from the gate semantics (I keeps, $V = H$ splits $\frac{1}{2}/\frac{1}{2}$, D and E reset), in `scaling/absorption.py`, at cost $O(4^N)$. It is a small-N diagnostic, not a sweep.

```
python -m qca_fragmentation.scaling.absorption --table
pytest qca_fragmentation/tests/test_absorption.py     # 58 tests
```

The tests pin every number in the table of §3.3, and check both directions of Proposition 3 as well as the double stochasticity of Proposition 1 — the latter holding for the unitary rules and failing for the dissipative ones.

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